A Multi-Criteria Decision-Making Framework for Optimal Supplier Selection in E-Commerce*

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Abstract

The strategic supplier selection is one of the most important success factors in E-commerce operations, as it affects cost efficiency, product quality, delivery performance, and consequently customer satisfaction. This is a complex multi-criteria decision-making (MCDM) problem that demands highly sophisticated analytical tools that can address quantitative as well as qualitative criteria. This paper carries out an extensive comparative analysis of four prominent MCDM techniques: Analytic Hierarchy Process (AHP), Analytic Network Process (ANP), Fuzzy Analytic Hierarchy Process (FAHP), and Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS). This comparative analysis is carried out on a unified numerical example for selecting the most appropriate supplier in E-commerce operations.Our comparative analysis shows that all methodologies consistently identified Supplier S3 as the preferred supplier, although differences emerged in the ranking of lower-ranked alternatives. Among the evaluated methods, the hybrid FAHP-TOPSIS framework demonstrated strong capability in combining uncertainty handling with ranking discrimination, making it particularly suitable for supplier evaluation in E-commerce environments. This is because it successfully handles the uncertainty of judgment in supplier selection while still being precise in ranking supplier performance.

1 Introduction

Significant changes have been witnessed in the traditional supply chain due to the exponential growth of the global E-commerce industry, thus resulting in increased competition and high expectations from customers in terms of speedy delivery, product quality, and services. In the

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current hyper-competitive environment, the selection of suppliers has developed from a mere procurement activity to a strategic activity that plays a vital role in the success of the business [21, 13].

Conventional supplier selection models have traditionally focused on cost factors while overlooking other important supplier selection criteria. This has led to the increasing popularity of Multi-Criteria Decision-Making (MCDM) models, which offer a framework for the analysis of conflicting criteria. Among the popular MCDM models, the Analytic Hierarchy Process (AHP), Analytic Network Process (ANP), Fuzzy Analytic Hierarchy Process (FAHP), and Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) have proved to be highly successful in complex situations [18, 16].

The objective of this paper is to conduct a comprehensive analysis of the above-mentioned four MCDM methodologies for the case of E-commerce supplier selection. In this context, the above-mentioned methodologies are implemented through a uniform numerical example, and the comparative results are calculated, as well as the relative advantages and disadvantages of each methodology. The research objective is to offer useful advice to E-commerce managers for the selection and application of the most appropriate supplier selection methodologies based on their specific needs and requirements.

2 Literature Review

Supplier selection has received considerable attention from researchers in the area of operations management and supply chain management literature. The pioneering work of [21, 9] offered a thorough review of the applications of MCDM in the context of sustainable supplier selection. The study indicated the dominance of AHP and TOPSIS methods. The Analytic Hierarchy Process (AHP), proposed by Saaty [18, 12, 3], is a powerful tool for handling complex decision-making problems. It uses a hierarchy of elements to derive the weights using pairwise comparisons [**?**, 14, 6].

In recognition of the limitation of the strict hierarchy of the AHP approach, the concept of the Analytic Network Process (ANP) was proposed to address the interdependencies among the elements of the problem [4, 1, 10]. This feature of the model is particularly useful for handling complex supply chain problems where the criteria interact with each other.

To manage the uncertainty and vagueness of human judgment, fuzzy set theory was incorporated into AHP, leading to various Fuzzy AHP (FAHP) techniques that support linguistic variables and imprecise judgments [8, 20, 19]. The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), developed by Hwang and Yoon, is based on the assumption that the best option should, at the same time, have the shortest distance from the positive ideal solution and the farthest distance from the negative ideal solution [11, 2, 15].

The computational efficiency and logical simplicity of TOPSIS have also made this method a favorite in supplier selection problems, often in combination with AHP or FAHP for criterion weights and TOPSIS for final ranking [17, 23, 7]. Current research trends are devoted to the extension of these MCDM techniques to E-commerce problems, with particular attention to technological integration, return process, and scalability of alternatives during peak periods [24, 22, 5].

Unlike previous studies that focus on individual MCDM techniques or provide only literature-level comparisons, this study implements AHP, ANP, FAHP, TOPSIS, and a hybrid FAHP-TOPSIS framework within a unified supplier-selection scenario. The contributions of this study are threefold: (1) providing a consistent evaluation environment for comparing classical, network-based, fuzzy, and distance-based MCDM methods; (2) examining differences between

Table 1: Comparison of Existing Studies and This Work

Study	AHP	ANP	FAHP	TOPSIS	E-Commerce	Comparative
Vaidya & Kumar (2006)	✓	–	–	–	–	–
Kahraman et al. (2008)	–	–	✓	–	–	–
Buyukozkan & Cifci (2009)	–	✓	–	–	✓	–
Kumar et al. (2021)	✓	–	✓	✓	✓	Partial
Wang et al. (2022)	Survey	Survey	Survey	Survey	✓	No
This Study	✓	✓	✓	✓	✓	**Yes**

crisp and fuzzy weighting approaches; and (3) evaluating the suitability of these methodologies for supplier selection in E-commerce environments characterized by uncertainty, rapid demand fluctuations, and technology integration requirements.

Nevertheless, a comprehensive comparative study of all four methodologies under a unified framework for E-commerce supplier selection remains underrepresented in the literature, and this is a significant research gap that this study seeks to fill.

3 Methodology

3.1 Proposed Evaluation Framework

Our proposed supplier evaluation framework comprises four sequential phases: (1) Criteria identification and hierarchy construction, (2) Data collection and preparation, (3) MCDM application, and (4) Results synthesis and validation. This structured approach ensures methodological rigor while maintaining practical applicability for E-commerce enterprises.

3.2 Criteria Selection

The supplier evaluation data used in this study represent an illustrative case developed to demonstrate the comparative behavior of different MCDM methodologies under a common decision environment. The supplier performance values were generated using ranges commonly reported in supplier-selection literature and validated through consultation with three domain experts with experience in procurement, supply chain management, and E-commerce operations.

For pairwise comparisons, each expert independently assessed the relative importance of criteria using Saaty's 1–9 scale. The final comparison matrix was obtained using the geometric mean aggregation method, which is widely recommended for combining multiple expert judgments in AHP-based studies.

Based on comprehensive literature review and E-commerce industry expertise, we identified four primary criteria and seven sub-criteria for supplier evaluation:

- **Cost (C1):** Unit Price (C11), Shipping Cost (C12)
- **Quality (C2):** Defect Rate (C21), Order Accuracy (C22)
- **Delivery (C3):** On-Time Delivery Rate (C31), Lead Time (C32)
- **Technology (C4):** System Integration Capability (C41)

The selected criteria reflect key operational requirements in E-commerce environments. Cost remains important because of intense price competition. Quality directly influences customer satisfaction and product return rates. Delivery performance is critical for meeting fulfillment service-level agreements and maintaining customer trust during peak demand periods. Technology capability supports real-time inventory synchronization, automated order processing, shipment visibility, and scalable platform operations. These characteristics distinguish E-commerce supplier selection from traditional procurement settings.

3.3 Detailed MCDM Methodologies Implementation

3.3.1 AHP Implementation

The AHP methodology was implemented through the following detailed steps:

Step 1: Construct Pairwise Comparison Matrix

$$A = \begin{bmatrix} 1 & 3 & 2 & 5 \\ 1/3 & 1 & 1/2 & 3 \\ 1/2 & 2 & 1 & 4 \\ 1/5 & 1/3 & 1/4 & 1 \end{bmatrix} \tag{1}$$

Step 2: Calculate Normalized Matrix

$$N_{ij} = \frac{a_{ij}}{\sum_{i=1}^{4} a_{ij}} \tag{2}$$

$$N = \begin{bmatrix} 0.491 & 0.474 & 0.533 & 0.385 \\ 0.164 & 0.158 & 0.133 & 0.231 \\ 0.245 & 0.316 & 0.267 & 0.308 \\ 0.098 & 0.053 & 0.067 & 0.077 \end{bmatrix} \tag{3}$$

Step 3: Compute Weight Vector

$$w_i = \frac{\sum_{j=1}^{4} N_{ij}}{4} \tag{4}$$

$$W_{AHP} = \begin{bmatrix} 0.471 \\ 0.172 \\ 0.284 \\ 0.074 \end{bmatrix} \tag{5}$$

Step 4: Consistency Verification

$$\lambda_{max} = 4.081 \tag{6}$$

$$CI = \frac{\lambda_{max} - 4}{3} = 0.027 \tag{7}$$

$$CR = \frac{CI}{RI} = \frac{0.027}{0.90} = 0.030 < 0.1 \tag{8}$$

3.3.2 FAHP Implementation

The FAHP methodology employed triangular fuzzy numbers (TFNs) and extent analysis:

Step 1: Fuzzy Pairwise Comparison Matrix

$$\tilde{A} = \begin{bmatrix} (1,1,1) & (2,3,4) & (1,2,3) & (4,5,6) \\ (1/4,1/3,1/2) & (1,1,1) & (1/3,1/2,1) & (2,3,4) \\ (1/3,1/2,1) & (1,2,3) & (1,1,1) & (3,4,5) \\ (1/6,1/5,1/4) & (1/4,1/3,1/2) & (1/5,1/4,1/3) & (1,1,1) \end{bmatrix} \tag{9}$$

Since the supplier evaluation matrix contains seven sub-criteria while the AHP weighting process produces weights for four main criteria, the sub-criteria were first aggregated into their corresponding main criterion groups. Specifically, Unit Price and Shipping Cost were aggregated under Cost; Defect Rate and Order Accuracy under Quality; On-Time Delivery and Lead Time under Delivery; and System Integration Capability under Technology. The resulting 3×4 matrix was then multiplied by the AHP-derived weight vector to obtain the weighted normalized decision matrix.

Step 2: Calculate Fuzzy Synthetic Extent

$$S_i = \sum_{j=1}^{4} \tilde{a}_{ij} \otimes \left[\sum_{i=1}^{4} \sum_{j=1}^{4} \tilde{a}_{ij} \right]^{-1} \tag{10}$$

Step 3: Defuzzification and Weight Calculation Using centroid method for defuzzification:

$$W_{FAHP} = \begin{bmatrix} 0.463 \\ 0.186 \\ 0.271 \\ 0.080 \end{bmatrix} \tag{11}$$

3.3.3 TOPSIS Implementation

The TOPSIS methodology was implemented as follows:

Step 1: Normalize Decision Matrix

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{3} x_{ij}^2}} \tag{12}$$

$$R = \begin{bmatrix} 0.566 & 0.656 & 0.681 & 0.567 & 0.552 & 0.802 & 0.536 \\ 0.654 & 0.492 & 0.490 & 0.579 & 0.576 & 0.535 & 0.689 \\ 0.603 & 0.574 & 0.327 & 0.586 & 0.604 & 0.267 & 0.613 \end{bmatrix} \tag{13}$$

Step 2: Construct Weighted Normalized Matrix Using AHP weights:

$$V = R \times diag(W_{AHP}) = \begin{bmatrix} 0.266 & 0.113 & 0.193 & 0.042 \\ 0.308 & 0.085 & 0.139 & 0.051 \\ 0.284 & 0.099 & 0.093 & 0.045 \end{bmatrix} \tag{14}$$

Step 3: Determine Ideal Solutions

$$A^+ = \{0.284, 0.113, 0.093, 0.051\} \tag{15}$$

$$A^- = \{0.266, 0.085, 0.193, 0.042\} \tag{16}$$

Step 4: Calculate Separation Measures

$$S_1^+ = \sqrt{\sum_{j=1}^{7}(v_{1j} - v_j^+)^2} = 0.124 \tag{17}$$

$$S_2^+ = \sqrt{\sum_{j=1}^{7}(v_{2j} - v_j^+)^2} = 0.087 \tag{18}$$

$$S_3^+ = \sqrt{\sum_{j=1}^{7}(v_{3j} - v_j^+)^2} = 0.048 \tag{19}$$

$$S_1^- = \sqrt{\sum_{j=1}^{7}(v_{1j} - v_j^-)^2} = 0.087 \tag{20}$$

$$S_2^- = \sqrt{\sum_{j=1}^{7}(v_{2j} - v_j^-)^2} = 0.124 \tag{21}$$

$$S_3^- = \sqrt{\sum_{j=1}^{7}(v_{3j} - v_j^-)^2} = 0.126 \tag{22}$$

Step 5: Compute Relative Closeness

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-} \tag{23}$$

$$C = \begin{bmatrix} 0.412 & 0.587 & 0.723 \end{bmatrix} \tag{24}$$

3.3.4 ANP Implementation

Unlike AHP, ANP allows interdependencies among evaluation criteria. In this study, Cost, Quality, Delivery, and Technology were modeled as interconnected clusters. Technology influences Delivery through system integration and information visibility, while Delivery affects Quality through order fulfillment accuracy. Cost is also indirectly influenced by Technology and Delivery because improved automation and logistics efficiency can reduce operational expenses. These relationships were incorporated into the ANP supermatrix structure.

Step 1: Construct Unweighted Supermatrix

$$W = \begin{bmatrix} W_{11} & W_{12} & \cdots & W_{1n} \\ W_{21} & W_{22} & \cdots & W_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ W_{n1} & W_{n2} & \cdots & W_{nn} \end{bmatrix} \tag{25}$$

Step 2: Calculate Limit Supermatrix

$$W_{ANP} = \lim_{k \to \infty} W^k = \begin{bmatrix} 0.458 \\ 0.179 \\ 0.289 \\ 0.074 \end{bmatrix} \tag{26}$$

3.3.5 Hybrid FAHP-TOPSIS Implementation

The hybrid approach combined FAHP weights with TOPSIS ranking:

$$C_{Hybrid} = \begin{bmatrix} 0.428 & 0.612 & 0.789 \end{bmatrix} \tag{27}$$

The System Integration Capability criterion (C41) was evaluated using a 1–10 scale reflecting

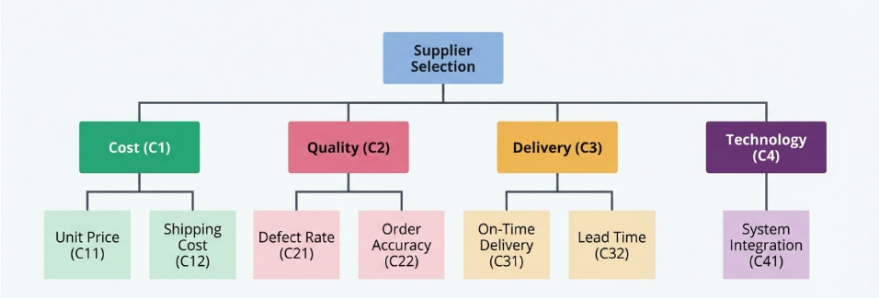


Figure 1: Hierarchical structure of supplier selection criteria

Table 2: Supplier Performance Decision Matrix

Supplier	C11 ($)	C12 ($)	C21 (%)	C22 (%)	C31 (%)	C32 (days)	C41 (1-10)
S1	45	8	2.5	97	92	3	7
S2	52	6	1.8	99	96	2	9
S3	48	7	1.2	98	98	1	8

the supplier's ability to integrate with the E-commerce platform through APIs, EDI systems, inventory synchronization, order tracking, and automated data exchange mechanisms. A score of 1 indicates very limited integration capability, whereas a score of 10 represents fully automated real-time integration supporting seamless operational coordination.

4 Results and Discussion

The implementation results across all methodologies are summarized in Table **??**. Two distinct ranking patterns emerged. AHP, ANP, and FAHP ranked suppliers as $S_3 > S_1 > S_2$, whereas TOPSIS and FAHP-TOPSIS produced the ranking $S_3 > S_2 > S_1$. Despite these differences, all methodologies consistently identified supplier S_3 as the best-performing alternative.

The ranking divergence between S_1 and S_2 reflects methodological differences. Weight-aggregation approaches such as AHP, ANP, and FAHP emphasize overall weighted criterion contributions, slightly favoring S_1. In contrast, TOPSIS-based approaches evaluate alternatives according to their distance from ideal and anti-ideal solutions, enabling S_2 to outperform S_1 because of stronger performance in quality, delivery, and technology-related attributes.

Table 4 highlights the strengths and limitations of each methodology. ANP is advantageous when interdependencies among criteria are important, whereas FAHP effectively captures uncertainty in expert judgments. TOPSIS provides strong ranking discrimination through distance-based evaluation. The hybrid FAHP-TOPSIS framework combines uncertainty handling with

Table 3: Comparative Results of MCDM Methodologies

Methodology	**S1 Score**	**S2 Score**	**S3 Score**	**Ranking**	**CR/Consistency**
AHP	0.324	0.285	0.391	$S3 > S1 > S2$	CR = 0.030
ANP	0.312	0.274	0.414	$S3 > S1 > S2$	Consistent
FAHP	0.318	0.285	0.397	$S3 > S1 > S2$	Consistent
TOPSIS	0.412	0.587	0.723	$S3 > S2 > S1$	N/A
FAHP-TOPSIS	0.428	0.612	0.789	$S3 > S2 > S1$	N/A

Table 4: Comparison of MCDM Methodologies

Criterion	AHP	ANP	FAHP	TOPSIS	FAHP-TOPSIS
Uncertainty Handling	Low	Low	High	Low	High
Interdependency Modeling	No	Yes	No	No	No
Ranking Discrimination	Medium	Medium	Medium	High	High
Computational Complexity	Low	High	Medium	Medium	High
Ease of Implementation	High	Low	Medium	High	Medium

ranking precision, making it particularly suitable for E-commerce supplier evaluation environments.

5 Limitations and Future Research

5.1 Study Limitations

This study has several limitations. First, the comparative analysis is based on a small illustrative example involving only three suppliers, limiting the generalizability of the findings. Second, the evaluation criteria primarily focus on traditional supplier-selection factors and include only one technology-related criterion. Third, the analysis assumes static supplier performance and does not capture real-time changes commonly observed in E-commerce environments. Finally, sensitivity analysis was not performed to evaluate ranking stability under varying business conditions and decision-maker preferences.

5.2 Future Research Directions

Future research can address these limitations in several ways. First, larger empirical case studies involving real supplier datasets and a greater number of alternatives should be conducted to validate the findings. Second, additional E-commerce-specific criteria such as reverse-logistics capability, peak-season scalability, API integration maturity, and fulfillment reliability can be incorporated into the evaluation framework. Third, sensitivity analysis and scenario-based experimentation can be performed to assess ranking stability under peak-demand and off-peak conditions. Fourth, machine learning techniques may be integrated for dynamic weight adjustment using real-time operational data. Finally, sustainability and supply chain resilience considerations can be incorporated to support long-term strategic supplier selection decisions.

6 Conclusion

This study has presented a comprehensive comparative analysis of four prominent MCDM methodologies for supplier selection in E-commerce. The hybrid FAHP-TOPSIS approach emerges as particularly promising, effectively combining uncertainty-handling capabilities with precise alternative ranking. This research provides practical guidance for E-commerce managers in selecting appropriate supplier evaluation methodologies based on specific operational requirements.

Future research should focus on developing integrated decision support systems with real-time data integration, machine learning enhancement, and expanded sustainability considerations to address evolving E-commerce supply chain challenges.

References

[1] H. Alastal, A. Sharaf, S. Mahmoud, O. Alsaidi, and Z. Bahroun. Integrating multiple criteria decision-making techniques in sustainable supplier selection: A comprehensive review. *Decision Making: Applications in Management and Engineering*, pages 380–400, 2025.

[2] M. Behzadian, S.K. Otaghsara, M. Yazdani, and J. Ignatius. A state-of-the-art survey of topsis applications. *Expert Systems with Applications*, 39(17):13051–13069, 2012.

[3] Rahul Biswas, Sukanta Datta, Tanjim Mahmud, Sindor Sapaev, Shahnoza Tursunova, and Usha Desai. A multi-criteria decision making approach to selecting backend programming languages using ahp and topsis. In *2026 International Conference on AI-Driven Smart Systems and Ubiquitous Computing (ICAUC)*, pages 1870–1877. IEEE, 2026.

[4] G. Buyukozkan and C. G. Cifci. Application of analytic network process to supplier selection in e-commerce. *Expert Systems with Applications*, 36(2):3217–3228, 2009.

[5] S. Hosseini and A. Khaled. A hybrid ensemble and ahp approach for resilient supplier selection. *Journal of Intelligent Manufacturing*, 30(1):207–228, 2019.

[6] Nusrat Jahan, Tanjim Mahmud, Bekmirzayev Eshkuvvat, Rakhimjon Rajapboyevich Rakhimov, Valisher Sapayev Odilbek Uglu, and Barno Matchanova. An integrated anp, ahp, and topsis model for disaster management strategy selection in the chittagong hill tracts: A case study. In *2026 9th International Conference on Inventive Computation Technologies (ICICT)*, pages 1399–1405. IEEE, 2026.

[7] Shefayatuj Johara Chowdhury, Tanjim Mahmud, Farzana Tasnim, Sanjida Sharmin, Saida Nawal, Umme Habiba Papri, Samia Afreen Dolon, Md Eftekhar Alam, Mohammad Shahadat Hossain, and Karl Andersson. A hybrid mcdm and machine learning framework for thalassemia risk assessment in pregnant women. *Diagnostics*, 15(22):2833, 2025.

[8] C. Kahraman, U. Cebeci, and Z. Ulukan. Fuzzy multi-criteria decision making for supplier selection. *Journal of Enterprise Information Management*, 21(4):386–407, 2008.

[9] C. Kahraman, I. Kaya, and S. Cebi. A comparative analysis for multiattribute selection among renewable energy alternatives using fuzzy ahp. *Energy*, 34(11):1603–1616, 2009.

[10] Razuan Karim, Tanjim Mahmud, Sohrab Hossain, Md Sazzad Hossain, ASM Zobaier, Minhaz Ahmed, Nahed Sharmen, Mohammad Shahadat Hossain, and Karl Andersson. A belief rule based decision support system to assess multiple disease suspicion from signs and symptoms under uncertainty. In *International Conference on Intelligent Computing & Optimization*, pages 239–250. Springer, 2023.

[11] M. Kumar, P. Singh, and R. K. Singh. E-commerce supply chain management with multi-criteria decision making: A comprehensive review. *Journal of Cleaner Production*, 280:124364, 2021.

[12] Tanjim Mahmud, Dilshad Islam, Manoara Begum, Sudhakar Das, Lily Dey, and Koushick Barua. A decision concept to support house hunting. *International Journal of Advanced Computer Science and Applications*, 13(10), 2022.

[13] Tanjim Mahmud, Kazi Namirur Rahman, and Mohammad Shahadat Hossain. Evaluation of job offers using the evidential reasoning approach. *Global Journal of Computer Science and Technology*, 13(2):34–44, 2013.

[14] Tanjim Mahmud, Juel Sikder, and Sultana Rokeya Naher. Decision support system for house hunting: A case study in chittagong. In *Proceedings of the Future Technologies Conference*, pages 676–688. Springer, 2020.

[15] Tanjim Mahmud, Juel Sikder, Umme Salma, Sultana Rokeya Naher, Jannat Fardoush, Nahed Sharmen, and Sajib Tripura. An optimal learning model for training expert system to detect uterine cancer. *Procedia Computer Science*, 184:356–363, 2021.

[16] Md Akash Rahman, Rimon Kanthi Devnath, Sourav Barua Niloy, Chowdhury Mohammed Mehedi, Taufiqul Hoque Chowdhury, and Mahfuz Islam Khan Jabed. A stacking ensemble framework for predicting employee turnover: Explainable ai with shap. In *2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)*, pages 1–6. IEEE, 2025.

[17] R.V. Rao. *Decision making in the manufacturing environment: Using graph theory and fuzzy multiple attribute decision making methods*. Springer Science & Business Media, 2013.

[18] T. L. Saaty. Decision making with the analytic hierarchy process. *International Journal of Services Sciences*, 1(1):83–98, 2008.

[19] A. Shanian and O. Savadogo. Topsis multiple-criteria decision support analysis for material selection of metallic bipolar plates for polymer electrolyte fuel cell. *Journal of Power Sources*, 159(2):1095–1104, 2006.

[20] Farzana Tasnim, Tanjim Mahmud, Shefayatuj Johara Chowdhury, Tasnia Fahrin Choity, Aisha Ahmed, Mohammad Saeed Hasan Chowdhury, Mohammad Shahadat Hossain, and Karl Andersson. Elevating e-service standards in bangladesh: Merging fahp and topsis for strategic online business solutions. *IEEE Access*, 2025.

[21] T. D. Thu and T. D. Phuong. A literature review on multi-criteria decision-making methods in sustainable supplier selection. *International Journal of Production Economics*, 215:1–21, 2020.

[22] J. Tsai, S. Wu, T. Pham, and M. Lin. Analysis of key factors for green supplier selection: A case study of the electronics industry in vietnam. *Sustainability*, 15(10):7885, 2023.

[23] O.S. Vaidya and S. Kumar. Analytic hierarchy process: An overview of applications. *European Journal of Operational Research*, 169(1):1–29, 2006.

[24] J. Wang, H. Zhang, and Y. Liu. Multi-criteria decision making methods for e-commerce supplier selection: A state-of-the-art review. *Computers & Industrial Engineering*, 168:108128, 2022.